



Summer Internship Task 2 — Offensive Security (Web Security Testing Framework)

Company: ITSOLERA PVT LTD

Task Title: Custom Web Security Testing Framework

Duration: 10 Days

Objective

Develop a modular Web Security Testing Framework capable of automating the detection of common web application security vulnerabilities against intentionally vulnerable applications.

The framework should simulate a simplified penetration testing workflow while maintaining clean architecture, extensibility, and professional reporting.

Task Description

1. Framework Development

Develop a CLI-based application using **Python** (preferred) or **Go**.

The framework should allow users to select different security testing modules independently.

Example:

```
python framework.py --target http://testphp.vulnweb.com --module xss
```

or

```
python framework.py --target http://demo.testfire.net --module headers
```

Required Modules

Module 1 – Security Headers Analyzer

Check whether the target website properly implements security headers.

Analyze:

- Content-Security-Policy
- X-Frame-Options
- X-Content-Type-Options
- Referrer-Policy
- Strict-Transport-Security
- Permissions-Policy

Display:

- Existing headers
 - Missing headers
 - Severity
 - Recommendations
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Module 2 – Authentication Assessment

Evaluate login functionality.

Checks may include:

- Weak password policy
 - Username enumeration
 - Login response differences
 - Account lockout detection
 - Session cookie analysis
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Module 3 – XSS Testing

Test discovered parameters for reflected XSS using safe payloads.

Requirements:

- GET parameters
 - POST parameters
 - Basic payload encoding
 - Detection based on reflected responses
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Module 4 – SQL Injection Testing

Perform safe SQL Injection checks.

Detect:

- SQL error messages
- Boolean-based response differences
- Time-based detection (optional)

Do **not** perform destructive exploitation.

Module 5 – Information Disclosure

Detect exposed resources including:

- robots.txt
 - sitemap.xml
 - .git
 - backup files
 - phpinfo
 - directory listing
 - exposed configuration files
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Reporting

Generate a professional report containing:

- Target URL
- Scan Date
- Modules Executed
- Vulnerabilities Found
- Severity

- Evidence
- Remediation
- Overall Risk Rating

Support at least one output format:

- HTML
- JSON

Project Structure

Organize your project professionally.

Example:

framework/

```
— framework.py
— modules/
  — headers.py
  — auth.py
  — xss.py
  — sqli.py
  — disclosure.py
— reports/
— utils/
— requirements.txt
— README.md
```

Advanced Features (Choose Any Three)

- Multi-threading
- Progress bar
- Custom User-Agent
- Cookie support
- Custom request headers
- Proxy support
- Timeout handling
- Colored terminal output
- HTML report generation
- JSON export
- Configuration file support

Bonus Features

Extra credit will be awarded for implementing any of the following:

- Burp Suite proxy integration
 - Session management
 - Login automation
 - JWT detection
 - API endpoint testing
 - Screenshot capture
 - Plugin architecture for adding future modules
 - Docker support
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Deliverables

Submit the following:

- Source Code
 - README.md
 - requirements.txt
 - Sample Report
 - GitHub Repository
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Rules

- Test **only** on intentionally vulnerable applications or systems that you own or have explicit authorization to assess.
- Recommended practice targets include:
 - DVWA
 - OWASP Juice Shop
 - bWAPP
 - WebGoat
 - Mutillidae
- Any copied project without proper attribution will be rejected.
- Write clean, modular, and well-documented code.

Submission

Upload the following before the deadline:

- GitHub Repository Link
- Source Code (.zip if GitHub is unavailable)
- Sample Report

Deadline: *24th July 2026*

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